# Time series: Exercises on ARMA properties

Andrey Pepelyshev

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The homework is provided on the last slide

# Representation of ARMA as MA

Task. Check that the ARMA process $x_t = 0.5x_{t-1} + \epsilon_t - 0.3\epsilon_{t-1}$ is causal and represent it as an MA process.

For the given process, we have $a(z) = 1 - 0.5z$. The root is $z_1 = 2$, which is located outside the unit circle, i.e. $|z_1| > 1$. Thus, the process $x_t$ is causal.

Since $(1 - 0.5B)x_t = (1 - 0.3B)\epsilon_t$ and $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \ldots$, we have

$$x_t = \frac{1 - 0.3B}{1 - 0.5B}\epsilon_t = (1 - 0.3B)(1 + 0.5B + 0.5^2B^2 + 0.5^3B^3 + \ldots)\epsilon_t$$

$$= (1 - 0.3B)\sum_{j=0}^{\infty} 0.5^j B^j \epsilon_t = \epsilon_t + \sum_{j=1}^{\infty}(0.5^j - 0.3 \cdot 0.5^{j-1})B^j\epsilon_t$$

$$= \epsilon_t + \sum_{j=1}^{\infty}(0.5 - 0.3) \cdot 0.5^{j-1}B^j\epsilon_t = \epsilon_t + \sum_{j=1}^{\infty} 0.2 \cdot 0.5^{j-1}\epsilon_{t-j}.$$

# Covariance function of ARMA via MA form

Task. Compute the covariance function of the ARMA process $x_t = 0.5x_{t-1} + \epsilon_t - 0.3\epsilon_{t-1}$ via the MA representation.

In slide 2, we obtained that $x_t$ has the form $x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$, where $c_0 = 1$ and $c_j = 0.2 \cdot 0.5^{j-1}$ for $j = 1, 2, \ldots$.

Then we have

$$
\begin{aligned}
R(k) &= \sigma^2 \sum_{s=0}^{\infty} c_s c_{s+|k|} \\
&= \sigma^2 [0.2 \cdot 0.5^{|k|-1} + \sum_{s=1}^{\infty} 0.2 \cdot 0.5^{s-1} 0.2 \cdot 0.5^{s+|k|-1}] \\
&= 0.2 \cdot 0.5^{|k|-1} \sigma^2 [1 + 0.2 \sum_{s=1}^{\infty} 0.5^{2s-1-1+1}] \\
&= 0.2 \cdot 0.5^{|k|-1} \sigma^2 [1 + 0.2 \frac{0.5}{1-0.25}] = \frac{0.85 \cdot 0.2}{0.75} 0.5^{|k|-1} \sigma^2.
\end{aligned}
$$

# Representation of ARMA as AR

Task. Check that the ARMA process $x_t = 0.5x_{t-1} + \epsilon_t - 0.3\epsilon_{t-1}$ is invertible and represent it as an AR process.

For the given process, we have $b(z) = 1 - 0.3z$. The root is $z_1 = 10/3$, which is located outside the unit circle, i.e. $|z_1| > 1$. Thus, this process is invertible.

Since $(1 - 0.5B)x_t = (1 - 0.3B)\epsilon_t$ and $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \ldots$, we have

$$\epsilon_t = \frac{1 - 0.5B}{1 - 0.3B} x_t = (1-0.5B)(1+0.3B+0.3^2B^2+0.3^3B^3+\ldots)x_t$$

$$= (1 - 0.5B)\sum_{j=0}^{\infty} 0.3^j B^j x_t = x_t + \sum_{j=1}^{\infty}(0.3^j - 0.5 \cdot 0.3^{j-1})B^j x_t$$

$$= x_t + \sum_{j=1}^{\infty}(0.3 - 0.5) \cdot 0.3^{j-1} B^j x_t = x_t - \sum_{j=1}^{\infty} 0.2 \cdot 0.3^{j-1} x_{t-j}.$$

# Checking causality and invertibility, ARMA(1,1)

Task. Identify if an ARMA(1,1) process $x_t - 1.5x_{t-1} = \epsilon_t + 0.5\epsilon_{t-1}$ is causal or not. Why?

For this process, the polynomial $a(z) = 1 - 1.5z$ has the root $z_1 = 2/3$, which is located inside the unit circle, i.e. $|z_1| < 1$. Thus, the process $x_t$ is not causal.

Task. Identify if an ARMA(1,1) process $x_t - 0.3x_{t-1} = \epsilon_t + 1.8\epsilon_{t-1}$ is invertible or not. Why?

For this process, the polynomial $b(z) = 1 + 1.8z$ has the root $z_1 = -10/18$, which is located inside the unit circle, i.e. $|z_1| < 1$. Thus, the process $x_t$ is not invertible.

# Variance and correlation function of AR(2)

Task. Find the variance and the correlation function of the AR(2) process $x_t = \alpha(x_{t-1} - x_{t-2}) + \epsilon_t$.

The 1-st and 2-nd Yule-Walker equations are

$$\rho(1) = \alpha - \alpha\rho(1),$$
$$\rho(2) = \alpha\rho(1) - \alpha.$$

Therefore, $\rho(1) = \frac{\alpha}{1+\alpha}$ and $\rho(2) = \alpha\frac{\alpha}{1+\alpha} - \alpha = -\frac{\alpha}{1+\alpha}$. Then from $k$-th Yule-Walker equation, we have $\rho(k) = \alpha\rho(k-1) - \alpha\rho(k-2)$, $k = 3, 4, \ldots$. The variance is $\sigma_x^2 = R(0) = \mathbf{E}x_t x_t = \mathbf{E}[\alpha(x_{t-1} - x_{t-2}) + \epsilon_t][\alpha(x_{t-1} - x_{t-2}) + \epsilon_t]$
$= \alpha^2 R(0) + \alpha^2 R(0) + \sigma^2 - 2\alpha^2 R(1)$
$= \alpha^2 R(0)(2 - 2\rho(1)) + \sigma^2 = 2\alpha^2\sigma_x^2(1 - \rho(1)) + \sigma^2$. Thus, $\sigma_x^2 = \sigma^2 \frac{1}{1-2\alpha^2(1-\rho(1))} = \sigma^2 \frac{1}{1-2\alpha^2\frac{1}{1+\alpha}} = \sigma^2 \frac{1+\alpha}{1+\alpha-2\alpha^2}$.

# Covariance function and spectrum of MA(2)

Task. Find the covariance function and the spectral density of the MA(2) process $x_t = \epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2}$.

Direct calculation gives

$$R(0) = \mathbf{E}x_t x_t = \mathbf{E}(\epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2})(\epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2})$$
$$= \sigma^2 + \beta^2\sigma^2 + \gamma^2\sigma^2 = \sigma^2(1 + \beta^2 + \gamma^2)$$
$$R(1) = \mathbf{E}x_t x_{t-1} = \mathbf{E}(\epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2})(\epsilon_{t-1} + \beta\epsilon_{t-2} + \gamma\epsilon_{t-3})$$
$$= \beta\sigma^2 + \gamma\beta\sigma^2 = \sigma^2\beta(1 + \gamma)$$
$$R(2) = \mathbf{E}x_t x_{t-2} = \mathbf{E}(\epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2})(\epsilon_{t-2} + \beta\epsilon_{t-3} + \gamma\epsilon_{t-4})$$
$$= \gamma\sigma^2$$

Then $R(k) = 0$ for $k = 3, 4, \ldots$.

The spectral density is $f(\lambda) = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty} R(k)\cos(\lambda k)$

$$= \frac{\sigma^2}{2\pi}[1 + \beta^2 + \gamma^2 + 2\beta(1 + \gamma)\cos(\lambda) + 2\gamma\cos(2\lambda)].$$

# Covariance function for given spectrum

Task. Let $x_t$ be a weakly stationary process with discrete time and spectral density $f(\lambda) = \pi - |\lambda|$. Find the covariance function of this process.

As given on slide 18 of Lecture 3, the covariance function is computed by

$$R(k) = \int_{-\pi}^{\pi} \cos(k\lambda) f(\lambda) d\lambda.$$

Thus, $R(0) = \int_{-\pi}^{\pi} (\pi - |\lambda|) d\lambda = \pi^2$ and for $k > 0$ we have

$R(k) = \int_{-\pi}^{\pi} \cos(k\lambda)\pi d\lambda - \int_{-\pi}^{\pi} \cos(k\lambda)|\lambda| d\lambda$

$= 0 - 2\int_0^{\pi} \cos(k\lambda)\lambda d\lambda = -2/k \int_0^{\pi} \lambda d \sin(k\lambda)$

$= -2/k[\lambda \sin(k\lambda)|_0^{\pi} - \int_0^{\pi} \sin(k\lambda) d\lambda]$

$= -2/k[0 + \frac{1}{k}\cos(k\lambda)|_0^{\pi}] = -2/k[0 + \frac{1}{k}((-1)^k - 1)]$

$= \frac{2(1-(-1)^k)}{k^2}$.

# Causality of AR(2) with given coefficients

Task. Identify the range for $\alpha$ when the AR(2) process $x_t = \alpha x_{t-1} + \alpha^2 x_{t-2} + \epsilon_t$ is causal.

The polynomial $a(z) = 1 - a_1 z - a_2 z^2$ with $a_1 = \alpha$ and $a_2 = \alpha^2$ has the roots $z_1 = \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{-2a_2}$, $z_2 = \frac{a_1 - \sqrt{a_1^2 + 4a_2}}{-2a_2}$.
Therefore,

$$z_1 = \frac{\alpha + \sqrt{\alpha^2 + 4\alpha^2}}{-2\alpha^2} = \frac{1 + \sqrt{5}}{-2\alpha}, \; z_2 = \frac{\alpha - \sqrt{\alpha^2 + 4\alpha^2}}{-2\alpha^2} = \frac{1 - \sqrt{5}}{-2\alpha},$$

which are real numbers (not complex). By Lemma, a process is causal if $|z_1| > 1$ and $|z_2| > 1$. We have $|\frac{1+\sqrt{5}}{-2\alpha}| > 1$ and $|\frac{1-\sqrt{5}}{-2\alpha}| > 1$. Thus, $x_t$ is causal if

$$\alpha \in \left( -\frac{\sqrt{5} - 1}{2}, \frac{\sqrt{5} - 1}{2} \right).$$

# R(0) and R(1) for ARMA(1,1)

Task. Find two values of the covariance function $R(0)$ and $R(1)$ for the ARMA(1,1) process $x_t = \alpha x_{t-1} + \epsilon_t + \beta\epsilon_{t-1}$.

First, we find
$R(0) = \mathbf{E}x_t x_t = \mathbf{E}(\alpha x_{t-1} + \epsilon_t + \beta\epsilon_{t-1})(\alpha x_{t-1} + \epsilon_t + \beta\epsilon_{t-1})$.
Expanding the brackets and using independence, we obtain

$$\begin{aligned} R(0) &= \alpha^2 R(0) + \sigma^2 + \beta^2\sigma^2 + 2\alpha\beta\mathbf{E}x_{t-1}\epsilon_{t-1} \\ &= \alpha^2 R(0) + \sigma^2 + \beta^2\sigma^2 + 2\alpha\beta\sigma^2. \end{aligned}$$

Thus, $R(0) = \sigma^2 \frac{1+\beta^2+2\alpha\beta}{1-\alpha^2}$.
Multiplying the given equation by $x_{t-1}$ and taking $\mathbf{E}$, we get

$$\mathbf{E}x_t x_{t-1} = \alpha\mathbf{E}x_{t-1}x_{t-1} + \mathbf{E}\epsilon_t x_{t-1} + \beta\mathbf{E}\epsilon_{t-1}x_{t-1},$$

that is, $R(1) = \alpha R(0) + 0 + \beta\sigma^2$.
Thus, $R(1) = \sigma^2(\alpha\frac{1+\beta^2+2\alpha\beta}{1-\alpha^2} + \beta)$.

# MA representation of AR(2)

Task. Compute the coefficients $c_0, c_1, c_2, c_3$ of the MA representation of the AR(2) process $x_t - x_{t-1} + 0.5x_{t-2} = \epsilon_t$.

Since the AR equation can be written as $(1-B+0.5B^2)x_t = \epsilon_t$ and $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \ldots$, we have

$$\begin{aligned} x_t &= \frac{1}{1-[B-0.5B^2]}\epsilon_t \\ &= (1 + [B-0.5B^2] + [B-0.5B^2]^2 + [B-0.5B^2]^3 + \ldots)\epsilon_t \end{aligned}$$

Since we are interested in terms with $B^j$, $j \leq 3$, we have

$$\begin{aligned} x_t &= (1 + B - 0.5B^2 + B^2 - 2 \cdot 0.5BB^2 + 0.5^2B^4 + B^3 + \ldots)\epsilon_t \\ &= \epsilon_t + \epsilon_{t-1} + 0.5\epsilon_{t-2} + 0 \cdot \epsilon_{t-3} + \ldots \end{aligned}$$

Thus, $c_0 = 1,\ c_1 = 1,\ c_2 = 0.5,\ c_3 = 0.$

# Covariance function of AR(3)

Task. Find the covariance function of the AR(3) process $x_t = \alpha x_{t-3} + \epsilon_t$.

Since the given equation can be written as $(1 - \alpha B^3)x_t = \epsilon_t$, we derive the following MA representation

$$x_t = \frac{1}{1 - \alpha B^3}\epsilon_t = \sum_{i=0}^{\infty} \alpha^i B^{3i} \epsilon_t = \sum_{i=0}^{\infty} \alpha^i \epsilon_{t-3i} = \sum_{j=0}^{\infty} c_j \epsilon_{t-j},$$

where $c_0 = 1$, $c_3 = \alpha$, $c_6 = \alpha^2$, $c_{3s} = \alpha^s$ and $c_j = 0$ for $j \notin \{0, 3, 6, 9, \ldots\}$. Since $R(k) = \sigma^2 \sum_{j=0}^{\infty} c_j c_{j+|k|}$, we have

$$R(k) = \begin{cases} r_{k/3}, & |k| \in \{0, 3, 6, 9, \ldots\} \\ 0, & |k| \notin \{0, 3, 6, 9, \ldots\} \end{cases}$$

where $r_s = \sigma^2 \sum_{j=0}^{\infty} \alpha^j \alpha^{j+|s|} = \sigma^2 \alpha^{|s|} \sum_{j=0}^{\infty} \alpha^{2j} = \sigma^2 \frac{\alpha^{|s|}}{1-\alpha^2}$.

# MA Representation of an AR(p) Process

Consider the AR($p$) model

$$x_t = a_1 x_{t-1} + a_2 x_{t-2} + \cdots + a_p x_{t-p} + \varepsilon_t,$$

with white noise $\varepsilon_t$. Its MA($\infty$) representation is

$$x_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j},$$

where the coefficients $\{\psi_j\}$ satisfy:

- **Initialization:**

$$c_0 = 1.$$

- **Recursion for $j \geq 1$:**

$$c_j = a_1 c_{j-1} + a_2 c_{j-2} + \cdots + a_p c_{j-p},$$

with $c_k = 0$ for $k < 0$.

# MA representation of AR(2)

Task. Compute the coefficients $c_0, c_1, c_2, c_3$ of the MA representation of the AR(2) process $x_t - x_{t-1} + 0.5x_{t-2} = \epsilon_t$.

**Initialization:** $c_0 = 1$.

**Recursion:** $c_j = a_1 c_{j-1} + a_2 c_{j-2}$, $j = 1, 2, \ldots$, with $c_{-1} = 0$.

$a_1 = 1$, $a_2 = -0.5$.

$c_1 = a_1 c_0 + a_2 c_{-1} = 1 \cdot 1 - 0.5 \cdot 0 = 1$

$c_2 = a_1 c_1 + a_2 c_0 = 1 \cdot 1 - 0.5 \cdot 1 = 0.5$

$c_3 = a_1 c_2 + a_2 c_1 = 1 \cdot 0.5 - 0.5 \cdot 1 = 0$

Thus, $c_0 = 1$, $c_1 = 1$, $c_2 = 0.5$, $c_3 = 0$.

# Homework: Learning with AI

Utilize AI tools to master the techniques of representation and covariance analysis for complex ARMA models.

- Provide AI with a specific ARMA(2,1) equation (e.g., $x_t - 0.8x_{t-1} + 0.15x_{t-2} = \epsilon_t + 0.3\epsilon_{t-1}$). Ask it to show the step-by-step expansion of the operator $(1 - 0.8B + 0.15B^2)^{-1}$ to find the first five $c_j$ coefficients.
- Ask AI to derive the analytical formula for $R(0)$ and $R(1)$ for the ARMA(1,1) process. Compare the AI's derivation with the results shown on Slide 10.
- Use AI to generate R-code that plots the roots of the characteristic polynomial $a(z)$ for the AR(2) task on Slide 11. Verify visually if they lie outside the unit circle.
- Ask AI to explain the relationship between the MA($\infty$) representation and the spectral density function. Request a plot showing how the spectrum of an AR(3) process [Slide 12] changes as $\alpha$ approaches the causality boundary.